Template

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2026-06-16

```r
library(tidyverse)
library(maps)

# Load data
gdp_data           <- read_csv("data/raw_gdp.csv")
unemployment_sex   <- read_csv("data/raw_unemployment_sex.csv")
unemployment_youth <- read_csv("data/raw_unemployment_youth.csv")

# Clean data
gdp_clean <- gdp_data %>%
  filter(geo == "Netherlands") %>%
  select(TIME_PERIOD, OBS_VALUE) %>%
  rename(Year = TIME_PERIOD, GDP_Growth = OBS_VALUE) %>%
  filter(!is.na(GDP_Growth))

unemp_clean <- unemployment_sex %>%
  filter(geo == "Netherlands") %>%
  select(TIME_PERIOD, OBS_VALUE) %>%
  rename(Year = TIME_PERIOD, Unemp_Rate = OBS_VALUE) %>%
  mutate(Unemp_Rate = ifelse(Unemp_Rate == -99, NA, Unemp_Rate)) %>%
  filter(!is.na(Unemp_Rate))

youth_clean <- unemployment_youth %>%
  filter(geo == "Netherlands") %>%
  select(TIME_PERIOD, OBS_VALUE) %>%
  rename(Year = TIME_PERIOD, Youth_Unemp_Rate = OBS_VALUE) %>%
  mutate(Youth_Unemp_Rate = ifelse(Youth_Unemp_Rate == -99, NA, Youth_Unemp_Rate)) %>%
  filter(!is.na(Youth_Unemp_Rate))

# Merge and engineer variables
combined_data <- gdp_clean %>%
  left_join(unemp_clean, by = "Year") %>%
  left_join(youth_clean, by = "Year") %>%
  mutate(Youth_Vulnerability_Ratio = Youth_Unemp_Rate / Unemp_Rate)
```

Set-up your environment

Title Page

A new sentence Include your names

Include the tutorial group number

Include your tutorial lecturer's name

Part 1 - Identify a Social Problem

Use APA referencing throughout your document. Here's a link to some explanation.

1.1 Describe the Social Problem

Include the following:

- Why is this relevant? (Smith, 2020)
- ...

Part 2 - Data Sourcing

2.1 Load in the data

Preferably from a URL, but if not, make sure to download the data and store it in a shared location that you can load the data in from. Do not store the data in a folder you include in the Github repository!

midwest is an example dataset included in the tidyverse package

2.2 Provide a short summary of the dataset(s)

In this case we see 28 variables, but we miss some information on what units they are in. We also don't know anything about the year/moment in which this data has been captured.

```
inline_code = TRUE
```

These are things that are usually included in the metadata of the dataset. For your project, you need to provide us with the information from your metadata that we need to understand your dataset of choice.

2.3 Describe the type of variables included

Think of things like:

- Do the variables contain health information or SES information?
- Have they been measured by interviewing individuals or is the data coming from administrative sources?

For the sake of this example, I will continue with the assignment...

Part 3 - Quantifying

3.1 Data cleaning

Our primary empirical data cleaning steps—including filtering exclusively for the Netherlands territorial boundaries and isolating the continuous percentage level transformations—were successfully executed within our master centralized data pipeline at the top of the file to preserve top-to-bottom reproducibility.

Please use a separate 'R block' of code for each type of cleaning. So, e.g. one for missing values, a new one for removing unnecessary variables etc.

3.2 Generate necessary variables

Variable 1: Youth Vulnerability Ratio

```r
# This variable was engineered during our master data pipeline at the top:
# combined_data <- combined_data %>% mutate(Youth_Vulnerability_Ratio = Youth_Unemp_Rate / Unemp_Rate)
# It standardizes the youth unemployment rate as a proportion of aggregate total unemployment.
```

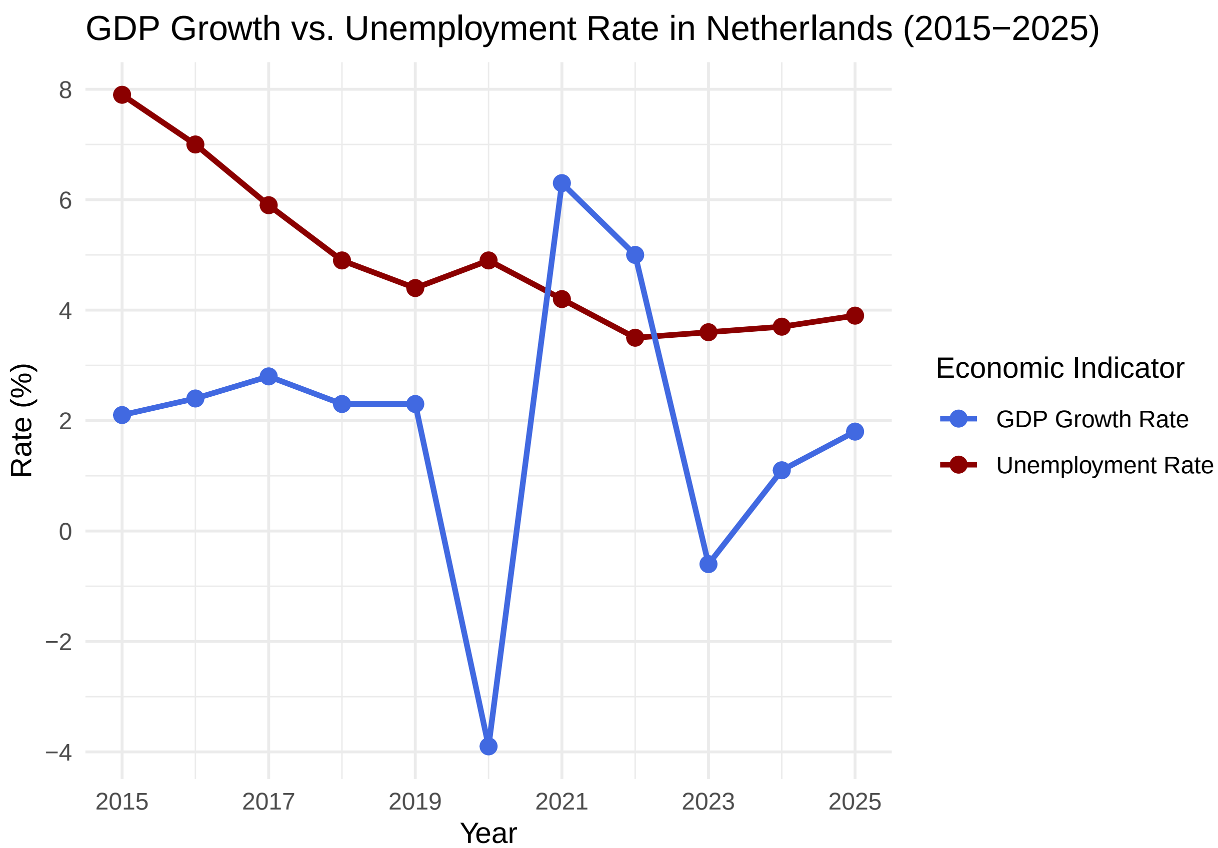
Variable 2

Relational database joins and chronological baseline alignments

were successfully completed using left_join() functions in the master pipeline.

3.3 Visualize temporal variation

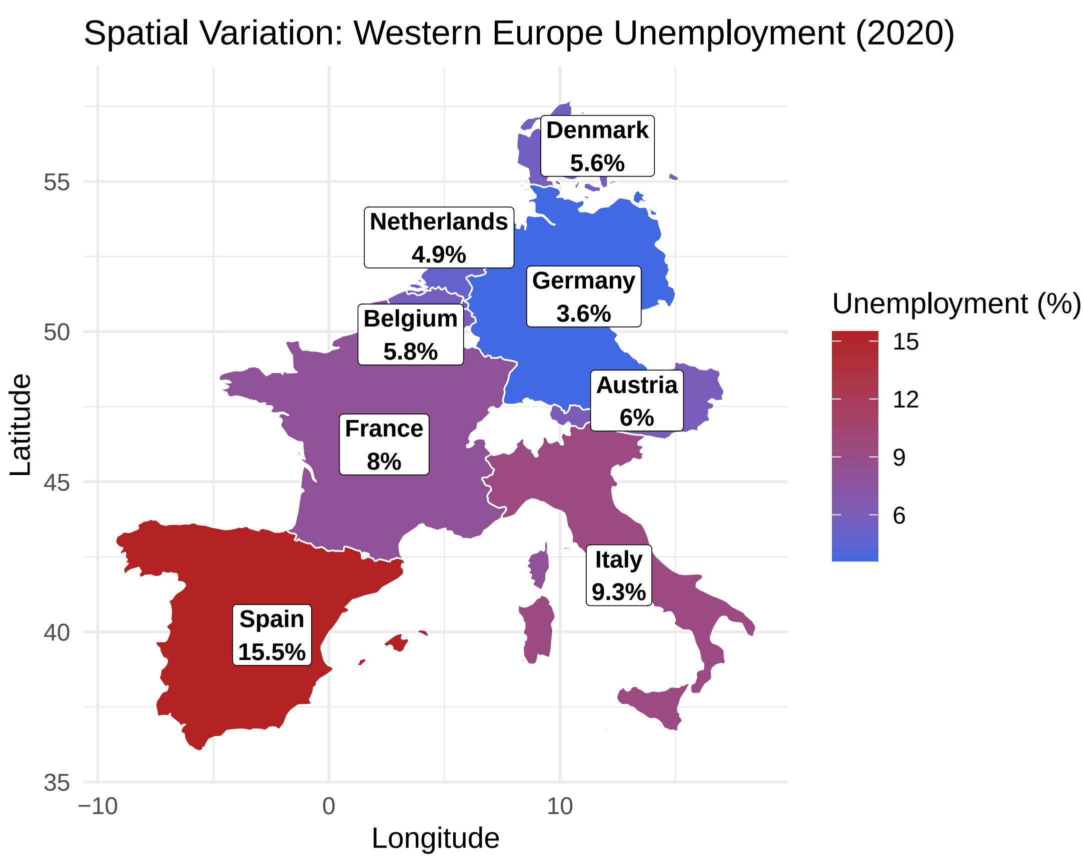
```r
ggplot(data = combined_data, aes(x = Year)) +
  geom_line(aes(y = Unemp_Rate, color = "Unemployment Rate"), linewidth = 1) +
  geom_point(aes(y = Unemp_Rate, color = "Unemployment Rate"), size = 2.5) +

  geom_line(aes(y = GDP_Growth, color = "GDP Growth Rate"), linewidth = 1) +
  geom_point(aes(y = GDP_Growth, color = "GDP Growth Rate"), size = 2.5) +

  scale_color_manual(values = c("Unemployment Rate" = "darkred", "GDP Growth Rate" = "royalblue")) +
  scale_x_continuous(breaks = seq(2015, 2025, by = 2)) +
  scale_y_continuous(breaks = seq(-4, 10, by = 2)) +
  theme_minimal() +
  labs(
    title = "GDP Growth vs. Unemployment Rate in Netherlands (2015-2025)",
    x = "Year",
    y = "Rate (%)",
    color = "Economic Indicator"
  )
```



3.4 Visualize spatial variation

```
europe_map <- map_data("world") %>%
  filter(region %in% c("Netherlands", "Germany", "Belgium", "France", "Italy", "Spain", "Austria", "Denm

europe_unemp_2020 <- unemployment_sex %>%
  filter(TIME_PERIOD == 2020) %>%
  rename(region = geo)

map_data_joined <- left_join(europe_map, europe_unemp_2020, by = "region")

country_labels <- map_data_joined %>%
  group_by(region) %>%
  summarise(
    long = mean(range(long)),
    lat = mean(range(lat)),
    Unemp_Rate = first(OBS_VALUE)
  ) %>%
  mutate(
    lat = case_when(
      region == "Netherlands" ~ lat + 1.0,
      region == "Belgium" ~ lat - 0.6,
      TRUE ~ lat
    ),
```

```r
    long = case_when(
      region == "Netherlands" ~ long - 0.5,
      region == "Belgium" ~ long - 0.9,
      region == "Germany" ~ long + 0.6,
      TRUE ~ long
    )
  )

ggplot(map_data_joined, aes(x = long, y = lat, group = group, fill = OBS_VALUE)) +
  geom_polygon(color = "white", linewidth = 0.3) +
  scale_fill_gradient(low = "royalblue", high = "firebrick", na.value = "gray80", name = "Unemployment

  geom_label(data = country_labels,
             aes(x = long, y = lat, label = paste0(region, "\n", Unemp_Rate, "%")),
             inherit.aes = FALSE,
             fill = "white", color = "black", size = 3.0, fontface = "bold",
             label.padding = unit(0.2, "lines"), label.size = 0.15) +

  coord_fixed(1.3) +
  theme_minimal() +
  labs(title = "Spatial Variation: Western Europe Unemployment (2020)", x = "Longitude", y = "Latitude"
```

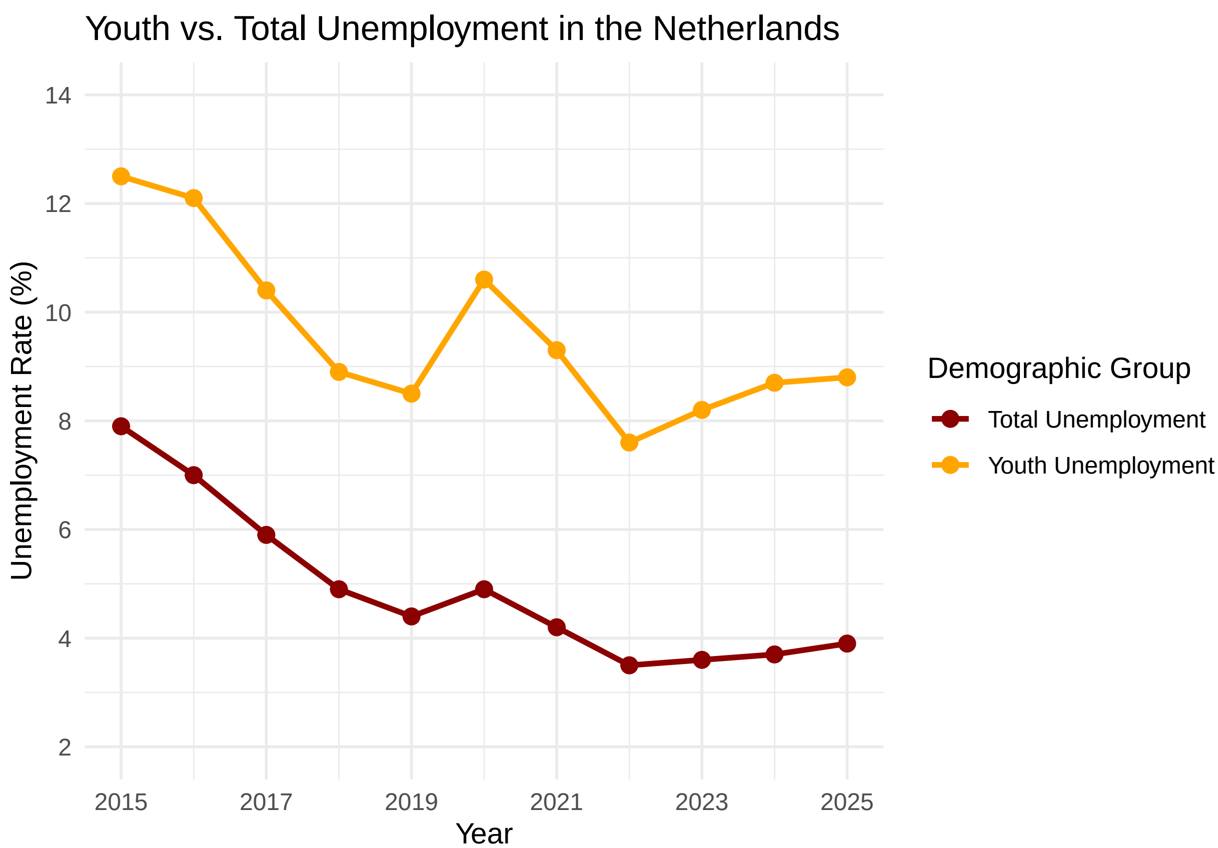


Here you provide a description of why the plot above is relevant to your specific social problem.

3.5 Visualize sub-population variation

What is the poverty rate by state?

```
df_total <- unemp_clean %>%
  rename(Unemployment_Rate = 2) %>%
  mutate(Demographic_Group = "Total Unemployment")

df_youth <- youth_clean %>%
  filter(Year >= 2015 & Year <= 2025) %>%
  rename(Unemployment_Rate = 2) %>%
  mutate(Demographic_Group = "Youth Unemployment")

youth_data <- bind_rows(df_total, df_youth)

ggplot(data = youth_data, aes(x = Year, y = Unemployment_Rate, color = Demographic_Group, group = Demog
  geom_line(linewidth = 1) +
  geom_point(size = 2.5) +
  scale_color_manual(values = c("Total Unemployment" = "darkred", "Youth Unemployment" = "orange")) +
  scale_x_continuous(breaks = seq(2015, 2025, by = 2)) +
  scale_y_continuous(limits = c(2, 14), breaks = seq(2, 14, by = 2)) +
  theme_minimal() +
  labs(
    title = "Youth vs. Total Unemployment in the Netherlands",
    x = "Year",
    y = "Unemployment Rate (%)",
    color = "Demographic Group"
  )
```

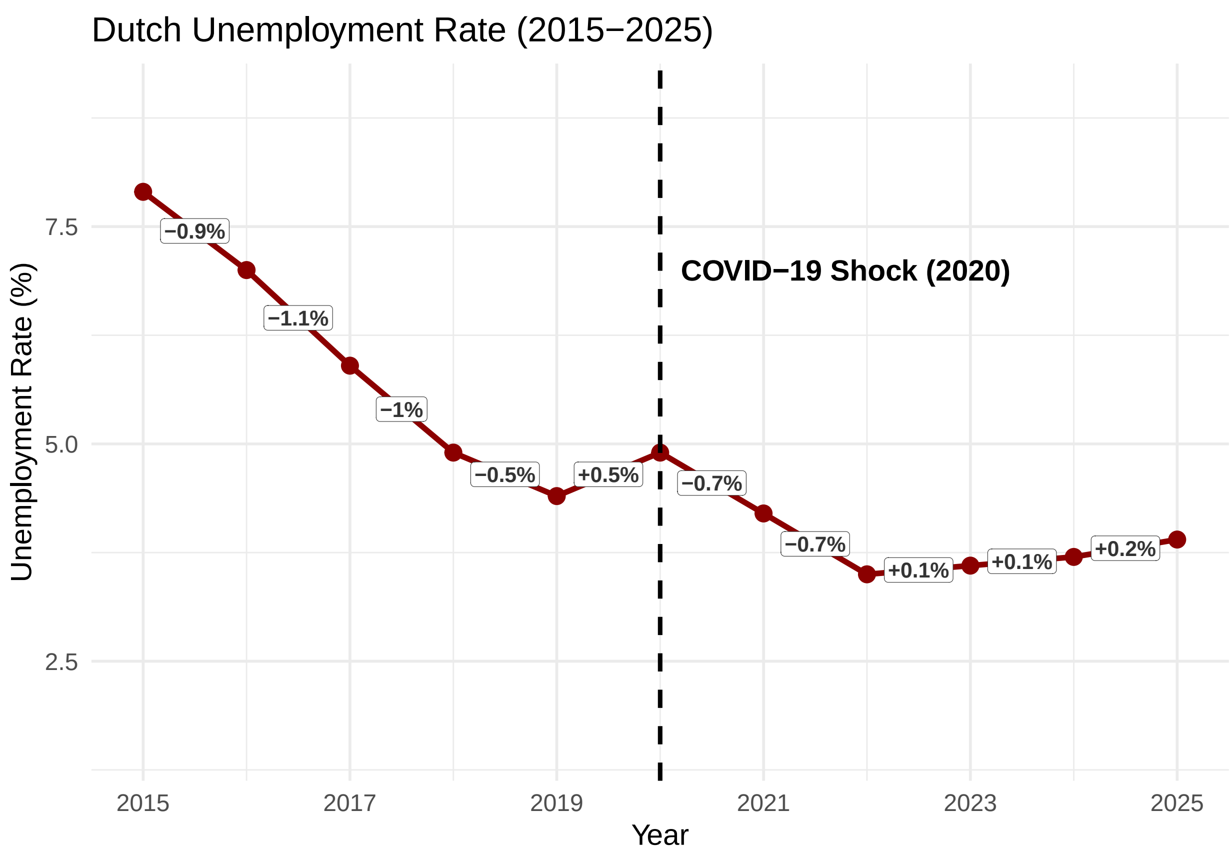


Here you provide a description of why the plot above is relevant to your specific social problem.

3.6 Event Analysis

```
temporal_changes <- combined_data %>%
  arrange(Year) %>%
  mutate(
    Ppt_Change = Unemp_Rate - lag(Unemp_Rate),
    x_mid = (Year + lag(Year)) / 2,
    y_mid = (Unemp_Rate + lag(Unemp_Rate)) / 2,
    Label_Text = ifelse(Ppt_Change > 0,
                        paste0("+", round(Ppt_Change, 1), "%"),
                        paste0(round(Ppt_Change, 1), "%"))
  ) %>%
  filter(!is.na(Ppt_Change))

ggplot(data = combined_data, aes(x = Year, y = Unemp_Rate)) +
  geom_line(color = "darkred", linewidth = 1) +
  geom_point(color = "darkred", size = 2.5) +
  geom_vline(xintercept = 2020, linetype = "dashed", color = "black", linewidth = 0.8) +
  annotate("text", x = 2020.2, y = 7, label = "COVID-19 Shock (2020)", hjust = 0, fontface = "bold") +

  geom_label(data = temporal_changes,
           aes(x = x_mid, y = y_mid, label = Label_Text),
```

```
            fill = "white", color = "gray20", size = 2.8, fontface = "bold",
            label.padding = unit(0.15, "lines"), label.size = 0.1) +

  scale_x_continuous(breaks = seq(2015, 2025, by = 2)) +
  theme_minimal() +
  labs(title = "Dutch Unemployment Rate (2015-2025)", x = "Year", y = "Unemployment Rate (%)") +
  ylim(1.5, 9)
```



Part 4 - Discussion

4.1 Discuss your findings

Part 5 - Reproducibility

5.1 Github repository link

Provide the link to your PUBLIC repository here: …

5.2 Reference list

Use APA referencing throughout your document.

Smith, J. (2020). My article. *Journal of Things*, *1*, 1–10.